

Dibayendu Mukherjee

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EDUCATION

Haldia Institute of Technology

Bachelor of Technology (B.Tech), Computer Science and Engineering

Current CGPA: 8.84/10 (Recent Semester: 9.09)

Expected July 2027

EXPERIENCE

Assistant Researcher

Haldia Institute of Technology

Present

Haldia, West Bengal

- Conducting advanced research in computer vision, digital watermarking, and deep learning systems architecture.
- Designing experimental PyTorch pipelines, implementing training stability fixes, and validating deep learning models for image forensics.
- Benchmarking model performance across rigorous diagnostic tests and documenting findings for academic publication.

Independent Researcher

Self-Directed

Jan 2024 – Dec 2025

Remote

- Researched and implemented deep learning and computer vision algorithms from first principles.
- Developed a custom autograd engine and handwritten digit classifier entirely in C.
- Designed and implemented Vision Transformers, CNNs, and optimization pipelines using PyTorch and TensorFlow.
- Built a custom SE-Net image classification model and explored reinforcement learning, transfer learning, and regularization techniques.

Technical & Strategy Intern

One Lab Ventures

Sept 2023 – Oct 2023

Remote

- Engineered automated data scraping and processing pipelines using Python, reducing manual data collection time by 80%.
- Automated internal reporting workflows and integrated CI/CD scripts, saving 15+ hours of manual operational work per week.
- Contributed to strategic product development initiatives, bridging technical implementation with business requirements.

TECHNICAL SKILLS

Languages: Python, C, C++, JavaScript, SQL

Machine Learning: PyTorch, TensorFlow, Computer Vision, ANNs, CNNs, RNNs, Transformers, Vision Transformers, Autoencoders, GANs, Reinforcement Learning, Sequence-to-Sequence Models, Image Segmentation, Object Detection, Metric Learning, Contrastive Learning, Transfer Learning

Backend & MLOps: Flask, REST APIs, Active Learning Pipelines, Online Learning Systems, Distributed Training (DDP)

Systems: WebAssembly, Emscripten, Tensor Engines, Autograd Systems, Computation Graphs

Tools: Git, GitHub, Linux, Docker, Hugging Face Hub

PROJECTS

Full-Stack AI Platform (HobbyFi)

REST APIs, CI/CD

- Architected scalable REST APIs and optimized database queries for improved backend performance.
- Implemented secure authentication, containerized deployment, and CI/CD workflows.
- Improved API latency and application scalability through backend optimization.

Plant Disease Classification Platform using SE-Net

TensorFlow, Flask

- Deployed a plant disease classification platform using a custom Squeeze-and-Excitation Network, categorizing 38 disease classes across 54,000 images.
- Established active learning and autonomous retraining pipelines, improving model validation accuracy to 95% over 50 training epochs.
- Streamlined automated model deployment via Hugging Face Hub, decoupling inference and training services for independent scaling.
- Optimized sample collection using confidence thresholds and mitigated catastrophic forgetting via experience replay buffers containing 500 historical samples.

Vision Transformer with Online Learning and Distributed Training

PyTorch, Flask

- Constructed a Vision Transformer from scratch in PyTorch, reducing training time by 50% using Distributed Data Parallel (DDP) across 2 GPUs, with architecture supporting multi-GPU scaling.
- Architected a Flask REST API for real-time image classification on Mini-ImageNet (approx. 100 classes).
- Formulated an online feedback pipeline enabling continual model adaptation, triggering incremental fine-tuning upon receiving 16 feedback samples.
- Automated zero-downtime model updates through dynamic weight reloading and asynchronous background retraining workflows, maintaining 100% service availability.

Vector Engine: Reverse-Mode Autograd Engine in C

C, Systems Engineering

- Authored a scalar-batched reverse-mode autograd engine in 1,200 lines of plain C, featuring dynamic computation graphs and tensor operations with broadcasting.
- Formulated a memory-efficient architecture using external gradient management and stateless tensors to minimize memory footprint.
- Implemented a custom SGD optimizer with momentum and batch-resizing capabilities for dynamic neural network training and inference.

WebAssembly Neural Network Playground

WebAssembly, JavaScript, Emscripten

- Developed an interactive neural network visualizer running the low-level C Vector Engine entirely in-browser via WebAssembly, bypassing Python dependencies.
- Bridged the C tensor engine with a reactive JavaScript frontend using Emscripten to enable real-time dynamic layer creation and training visualization.
- Engineered dynamic memory management within the Wasm bridge to prevent leaks during real-time graph rendering and model morphing.

Adversarial Dual-Stream Autoencoder (ADSA) for Forensic Tamper Detection

Research Project

- Designed a 4-stage forensic AI pipeline integrating semantic segmentation, dual-stream autoencoders, and embedding of 1,024 chaotic watermarks within a 28x28 latent space.
- Engineered a ResNet-34 based forensic extractor with FPN decoding and ArcFace metric learning for watermark identity recovery.
- Achieved Dice score of 0.9541 for tamper localization while maintaining watermark invisibility with an energy ratio below 0.00007671.
- Resolved critical training instability issues across 80 epochs through curriculum learning, BatchNorm stabilization, latent alignment, and high-frequency preservation techniques.

RESEARCH INTERESTS

Computer Vision, Digital Watermarking, Image Forensics, Representation Learning, Deep Learning Systems, Continual Learning, ML Systems Engineering

LEADERSHIP & ACTIVITIES

- **Workshop Organizer & Instructor**, AI: Developed and delivered technical workshops on Python, Machine Learning, AI fundamentals, and Agentic AI automation.
- Active contributor to independent research and open-source machine learning projects.